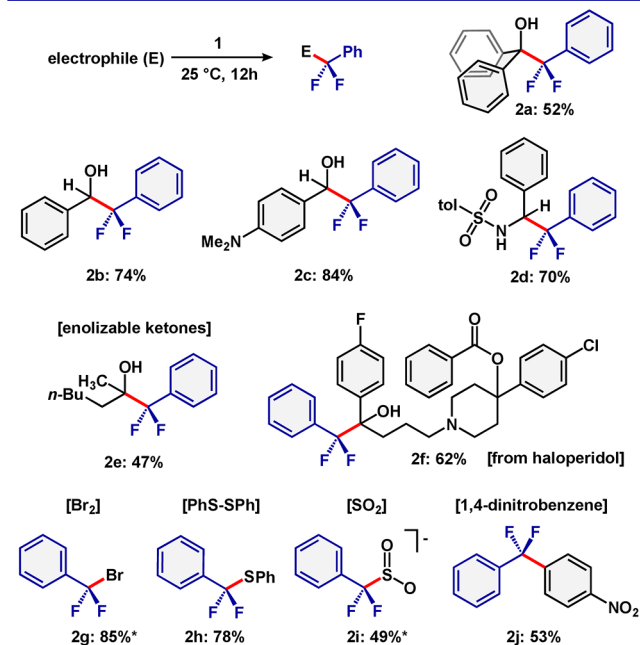
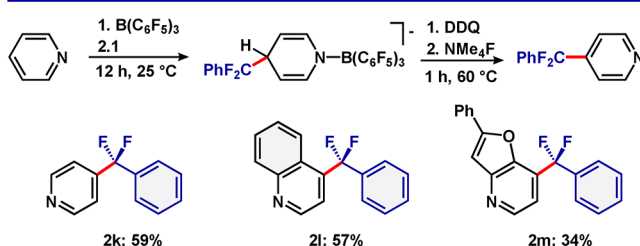


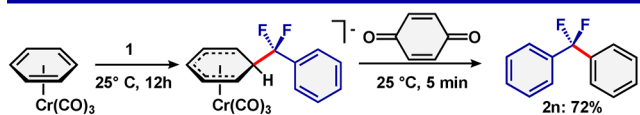
a) 1,2 addition, nucleophilic substitution



b) C-H functionalization of pyridines



c) chromium mediated benzene activation



d) palladium mediated cross-coupling with aryl halides

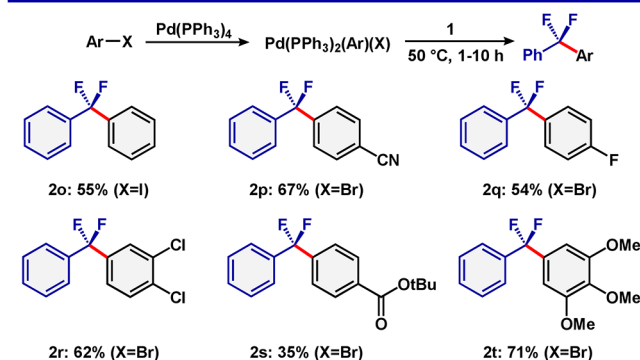


Figure 2. (a) Reactions with ketones, aldehydes, imines, disulfides, Br₂, SO₂, and 1,4-dinitrobenzene. (b) Two-phase C-H difluorobenzoylation of pyridines. (c) Chromium(0)(CO)₃ mediated benzene C-H difluorobenzoylation. (d) Palladium-mediated difluorobenzoylation.

thrombin inhibitor containing an Ar-CF₂-R linkage, significant effort was expended to avoid either of these routes and ultimately the molecule was built outward from bromodifluoroacetate.⁸

In contrast to the late-stage fluorination strategies described above, fragment coupling with [ArCF₂] units enables a more modular synthetic approach. However, electrophilic ArCF₂⁺ and radical ArCF₂[•] based disconnections present significant limitations: electrophilic ArCF₂Br starting materials are not widely available, radical cross-coupling between sulfone (R-CF₂-SO₂R)⁹ electrophiles and aryl zinc nucleophiles is limited to aliphatic RCF₂ transfer, and generation of Ar-CF₂[•] from Ar-CF₃¹⁰ requires strong reductants and the reported scope in radicalphile is limited. Compared to electrophilic and radical methodologies,¹¹ reactions with ArCF₂⁻ nucleophiles are underdeveloped despite the wide availability of C- electrophilic (Ar-X, carbonyl, imine) functional groups. ArCF₂SiMe₃ reagents are the only access point to this reactivity manifold, but synthetic routes to these reagents are highly limited in scope.^{5c,12}

In contrast to the poor availability of ArCF₂SiMe₃ pronucleophiles, difluoromethyl arenes (ArCF₂H) could be ideal masked ArCF₂⁻ nucleophiles upon deprotonation. Recent progress in transition metal catalyzed difluoromethylation has made these compounds accessible and structurally diverse starting materials.¹³ However, ArCF₂H groups have low acidity (no reaction with KN(SiMe₃)₂)¹⁴ and, if deprotonated, the revealed ArCF₂⁻ fragment is unstable to α-fluoride elimination.¹⁵ The use of compatible pairs of strong bases, which deprotonate ArCF₂H groups, and Lewis acids, which stabilize fluoroalkyl anions, may address these issues.¹⁶ Appropriate selection of a Lewis acid that can capture ArCF₂⁻ after ArCF₂H deprotonation, and later release ArCF₂⁻ in reactions with electrophiles could enable the broad-scope synthesis of a new class of nucleophilic difluorobenzoylation reagents.

Three key requirements must be met to transform ArCF₂H groups into useful ArCF₂⁻ synthons using Brønsted base/Lewis acid pairs: (1) the base used for C-H deprotonation must be sufficiently strong, (2) the Lewis acid must not irreversibly bind the Brønsted base (Figure 1c), and (3) the Lewis acid must be capable of both ArCF₂⁻ capture/stabilization and ArCF₂⁻ release to facilitate subsequent reactions with electrophiles. We hypothesized that PhCH₂⁻, a strong base, and hexamethylborazine (B₃N₃Me₆), a privileged Lewis acid for the stabilization of fluoroalkyl anions, could satisfy these criteria. Though pK_a measurements for ArCF₂-H bonds are unavailable, we anticipated that KCH₂Ph would be capable of deprotonating HCF₂Ph,¹⁵ satisfying criterion (1). This base (KCH₂Ph) forms a reversible adduct with B₃N₃Me₆, satisfying criterion (2). In silico studies revealed that B₃N₃Me₆ has a PhCF₂⁻ affinity of -23.3 kcal/mol, which is similar to CF₃⁻ (-22.6 kcal/mol). B₃N₃Me₆-CF₃⁻ adducts readily release CF₃⁻, suggesting that B₃N₃Me₆ could serve an appropriate Lewis acid for both stabilization and transfer of PhCF₂⁻, satisfying criterion (3).^{16a,b}

Addition of PhCF₂H to an equimolar combination of B₃N₃Me₆, KCH₂Ph, and 18-crown-6 in THF at 25 °C afforded [K(18-crown-6)][PhCF₂-B₃N₃Me₆] (1) in 98% chemical yield and 72% isolated yield (>9 g scale) as a crystalline solid (Figure 1d). Similar reactions carried out using KC(Me)Ph₂ (3%) and KCHPh₂ (0%) provided a significantly lower yield of 1, and allowed the pK_a of PhCF₂H to be bracketed between 41 (PhCH₃) and 35 (Ph₂CHMe). We also investigated the use of bulky diisopropylamide bases (Li, Na, and KN(iPr)₂) and found that KN(iPr)₂ afforded the product in high yield (91%), whereas Na- (21%) and LiN(iPr)₂ (0%) afforded lower yields of CF₂Ph⁻ containing product, consistent

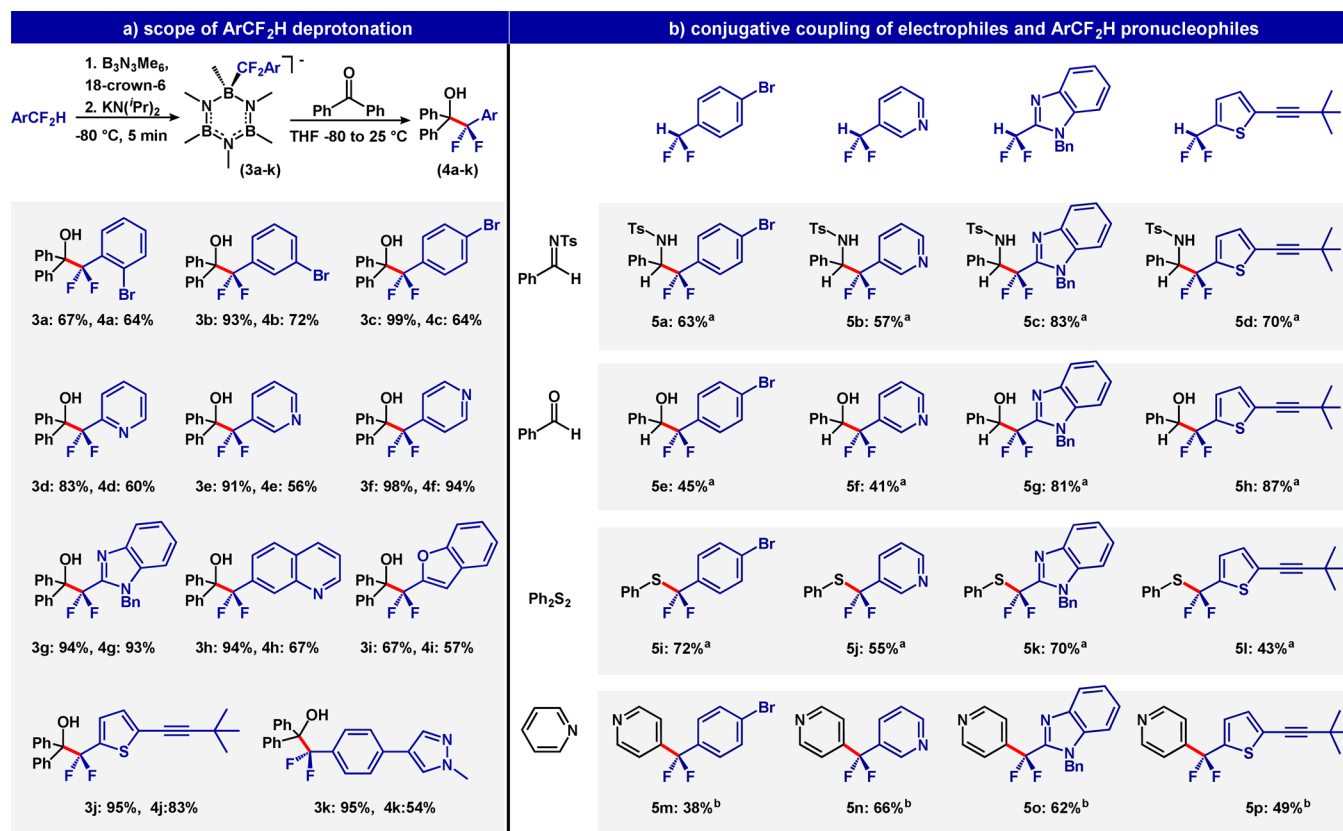


Figure 3. (a) Scope of ArCF₂H deprotonation, ArCF₂[−] stabilization by B₃N₃Me₆, and transfer of ArCF₂[−] equivalents. (b) Coupling a pool of four distinct ArCF₂[−] nucleophiles and a pool of four distinct electrophiles. ^a12 h, 25 °C. ^b1. Pyridine and B(C₆F₅)₃, 5 min, −80 °C; ArCF₂H, B₃N₃Me₆, 18-crown-6, and KN(iPr)₂, 10 min, −80 °C; 2. Pyridine–B(C₆F₅)₃ and ArCF₂–B₃N₃Me₆, 12 h, 25 °C; 3. DDQ, 15 min, 25 °C; 4. NMe₄F, 1 h, 60 °C (combined yield through five steps).

with α -defluorination by these more Lewis acidic counter-cations. Defluorination is not promoted by less Lewis acidic B₃N₃Me₆ or K(18-crown-6)⁺.

We experimentally evaluated the ability of **1** to transfer CF₂Ph[−] equivalents to electrophiles. **1** readily difluorobenzylates a variety of simple carbonyl and imine containing electrophiles (**2a–2d**, 52–84%) (Figure 2). These reactions proceeded to completion at room temperature in 12 h. Unlike the analogous CF₃[−] transfer reagent ([K(18-crown-6)][CF₃–B₃N₃Me₆]), **1** can transfer PhCF₂[−] to enolizable carbonyl compounds (**2e–2f**, 47% to 62%). The nucleophilic addition of ArCF₂[−] tolerates tertiary amines, tosyl groups, esters, and aromatic halides, and the functional group tolerance is highlighted by the selective difluorobenzylation of haloperidol benzoyl ester (**2f**), a derivatized antipsychotic. Sulfur dioxide also undergoes 1,2-addition to afford potassium difluorobenzylsulfinate (**2i**, 49%), which is an oxidatively activated PhCF₂[−] proradical capable of facilitating addition to α/β -unsaturated tosyl amides.¹⁷ In addition to 1,2-addition reactions, **1** can also difluorobenzylate diphenyldisulfide (Ph₂S₂) and bromine (Br₂) (**2h**, 78%; **2g**, 85%).

Doubly benzylic difluoromethylene linkages are especially difficult to synthesize using current methodologies. The only available approaches use ArCF₂Br starting materials, which must be synthesized through radical bromination or deoxy-fluorination strategies. Using **1**, dinitrobenzene can be directly difluorobenzylated in a net nucleophilic aromatic substitution (S_NAr) reaction (**2j**, 53%). When activated with a bulky and strong Lewis acid, B(C₆F₅)₃, pyridines also react with **1** to

generate a LA-stabilized anionic σ -adduct, which can be oxidized with DDQ and deprotected with NMe₄F to generate the aromatic 4-difluorobenzylated pyridine. These reactions can be conducted in one pot with a single solvent, affording the final product in good combined yield through four telescoped steps (**2k–2m**, 34–59%). An alternative activation strategy for nonheterocyclic arenes, such as benzene, is to use chromium(0) tricarbonyl. Treatment with **1** affords a stable σ -adduct that can be oxidized with benzoquinone to afford diphenyl difluoromethane (**2n**, 72%).

In addition to direct difluorobenzylation of activated arenes, we assessed the ability of **1** to access Ar–CF₂–Ar species from widely available aryl bromides and iodides through palladium-mediated cross coupling. Ar/ArCF₂ coupling reactions are currently limited to pathways in which ArCF₂[−] ligands are constructed through either:CF₂ insertion¹⁸ or ArCF₂Br oxidative addition,¹⁹ which present challenges for broad scope applications; notably, reactions using nucleophilic ArCF₂[−] reagents are unknown. We used Pd(PPh₃)₄ as a readily available stoichiometric reagent to mediate Ar–Br and Ar–I cross-coupling reactions with **1** as a transmetalation reagent. Treating *in situ* generated Pd(PPh₃)₂(Ar)X species with **1** afforded Ph–CF₂–Ar products (**2o–2t**, 35–71%), representing the first example of cross coupling using a nucleophilic ArCF₂[−] transmetalation reagent. This synthetic route provides a complementary strategy to access less electrophilic aryl halide coupling partners that are not amenable to direct nucleophilic addition reactions.

We next investigated the activation of difluoromethyl arenes beyond PhCF_2H . A wide variety of difluoromethyl (hetero)-arenes react with $\text{KN}(\text{iPr})_2/\text{B}_3\text{N}_3\text{Me}_6/18\text{-crown-6}$ in THF at -80°C to form stabilized ArCF_2^- synthons **3a–3k** in high yield (Figure 3a). When KCH_2Ph was used in place of $\text{KN}(\text{iPr})_2$, significantly lower yields were observed. Aryl bromides are not tolerated by electrochemical and photoredox benzylic fluorination methods,^{7,10,20} and are susceptible to *o*-metalation/benzyne formation. However, we accessed these nucleophilic ArCF_2^- units through deprotonation/capture reactions. 2- (**3a**, 67%), 3- (**3b**, 93%), and 4- (**3c**, 99%) bromobenzenes were all transformed into $\text{B}_3\text{N}_3\text{Me}_6/\text{ArCF}_2^-$ adducts in high yield. Difluoromethyl heteroarenes were also deprotonated to reveal ArCF_2^- synthons. In addition resonance-activated 2- and 4-difluoromethylpyridine (**3d**, 83%, **3f**, 98% chemical yield), 3-difluoromethylpyridine and 7-difluoromethylquinoline were deprotonated in high yield (**3e**, **3h**, 91%, 94%). Difluoromethylated five-membered heterocycles also undergo ArCF_2H deprotonation, with 2-difluoromethyl benzofuran (**3i**, 67%), *N*-benzyl-2-difluoromethyl-benzimidazole (**3g**, 94%), 1-difluoromethyl-4-(1-methyl-4-pyrazolyl)benzene (**3k**, 95%), and 2-difluoromethyl-5-(*tert*-butylacetylenyl)-thiophene (**3j**, 95%) providing access to $\text{B}_3\text{N}_3\text{Me}_6/\text{ArCF}_2^-$ adducts. The generated reagents efficiently transfer ArCF_2^- to Ph_2CO , affording the 1,2-addition products from ArCF_2H (**4a–4j**). Finally, a control reaction that omitted $\text{B}_3\text{N}_3\text{Me}_6$ from the reaction to form **4b** did not afford the desired product. This result illustrates the synergistic requirement of a compatible Lewis acid/base pair for both ArCF_2^- capture and transfer.

We demonstrated the robustness of this methodology by combining a pool of four structurally distinct difluoromethyl-(hetero)arenes with a pool of four representative organic electrophiles (Figure 3b; imine, aldehyde, disulfide, and LA-activated pyridine) to generate 16 unique products (**5a–5p**, 41–87%) over two steps. The reactions provide rapid access to structural diversity from simple precursors. Notably, several of the products contain first-in-class $\text{Ar}-\text{CF}_2-\text{R}$ linkages, such as imidazole- CF_2-SR (**5c**) and thiophene- CF_2 -carbamine (**5d**), structural cores that can be mapped onto biologically active nonfluorinated molecules.

In conclusion, we have developed a general approach for the conversion of common difluoromethyl (hetero)arenes into stabilized, structurally diverse, and previously inaccessible ArCF_2^- transfer reagents. The reagents are highly reactive, transferring ArCF_2^- to a wide variety of organic electrophiles to form ArCF_2-C bonds, and enable four new methods for the construction of $\text{Ar}-\text{CF}_2-\text{Ar}$ linkages. We anticipate that the use of difluoromethyl groups as sources of ArCF_2^- will enable the introduction of difluoromethylene linkages as easily accessible functional groups that will find applications in drug discovery.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/jacs.8b06093.

Crystallographic information for **1** (CIF)

Synthetic details, characterization (PDF)

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Notes

The authors declare the following competing financial interest(s): A patent application on this material has been submitted.

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